

Python — Chapter 1 & 2 Test

Total – 50 marks | duration 50 min

Part A — MCQs

30 Questions | 1 mark each

1. The step-by-step procedure used to solve a problem is called:

- A. Compiler
- B. Algorithm
- C. Interpreter
- D. Syntax

2. Which of the following is mainly used to represent an algorithm graphically?

- A. Compiler
- B. Flowchart
- C. Variable
- D. Interpreter

3. Which flowchart symbol is generally used for a decision or condition?

- A. Rectangle
- B. Oval
- C. Diamond
- D. Parallelogram

4. Which flowchart symbol is generally used for Start and End?

- A. Rectangle
- B. Diamond
- C. Oval/Terminal
- D. Parallelogram

5. Which flowchart symbol is used for processing/calculation?

- A. Rectangle
- B. Diamond
- C. Oval
- D. Arrow

6. Which flowchart symbol is commonly used for Input/Output?

- A. Rectangle
- B. Diamond
- C. Parallelogram
- D. Circle

7. Pseudocode is mainly used to:

- A. Execute a program
- B. Represent program logic in an understandable form
- C. Translate code into machine language
- D. Store data permanently

8. An algorithm should be:

- A. Confusing
- B. Ambiguous
- C. Clear and unambiguous
- D. Written only in Python

9. Which of the following is NOT a programming language?

- A. Python
- B. C
- C. Java
- D. Flowchart

10. Python is a:

- A. High-level programming language
- B. Machine language
- C. Flowcharting language
- D. Hardware language

11. Which of the following is a major advantage of an algorithm?

- A. It helps plan the solution before coding
- B. It automatically executes the program
- C. It replaces the computer
- D. It removes the need for testing

12. Which of the following is language-independent?

- A. Python program
- B. C program

- C. Algorithm
- D. Python syntax

13. A flowchart represents a solution using:

- A. Only English sentences
- B. Standard graphical symbols
- C. Only Python statements
- D. Machine instructions

14. Which of the following represents the normal sequence of problem solving?

- A. Code → Problem → Algorithm → Testing
- B. Problem → Algorithm → Code → Testing
- C. Testing → Code → Problem → Algorithm
- D. Algorithm → Testing → Problem → Code

15. Which component translates source code into a form suitable for execution by the computer?

- A. Compiler
- B. Flowchart
- C. Algorithm
- D. Pseudocode

16. Which program generally executes a program by translating instructions during execution?

- A. Compiler
- B. Interpreter
- C. Flowchart
- D. Editor

17. Finding and fixing errors in a program is called:

- A. Testing
- B. Debugging
- C. Compiling
- D. Designing

18. Which error occurs when the rules/syntax of a programming language are violated?

- A. Logical error
- B. Syntax error
- C. Hardware error
- D. Portability error

19. A program runs successfully but gives the wrong answer. This is most likely a:

- A. Syntax error
- B. Logical error
- C. Compilation error
- D. Typing error

20. Which of the following is an example of a sequence structure?

- A. if-else
- B. for loop
- C. Statements executed one after another
- D. Repeated execution based on a condition

21. Consider the following steps:

- 1. Start
- 2. Input A and B
- 3. Calculate $C = A + B$
- 4. Display C
- 5. Stop

Which step represents the **processing**?

- A. Step 1
- B. Step 2
- C. Step 3
- D. Step 4

22. Suppose an algorithm is written to calculate the area of a rectangle. Which information is required as input?

- A. Area only
- B. Length and breadth
- C. Perimeter only
- D. Output only

23. If a flowchart asks, "Is age ≥ 18 ?", which type of operation is being performed?

- A. Input
- B. Processing
- C. Decision
- D. Output

24. Which of the following is the best reason for creating an algorithm before writing code?

- A. It makes the computer faster
- B. It helps us plan the solution clearly
- C. It removes the need for programming
- D. It automatically creates the program

25. The same algorithm can be implemented using Python, C or Java because an algorithm is:

- A. Dependent on Python
- B. Dependent on hardware
- C. Independent of a particular programming language
- D. Written only using machine code

26. Which sequence correctly represents the development of a simple program?

- A. Understand problem → Design solution → Write code → Run/Test
- B. Write code → Understand problem → Design solution → Test
- C. Run → Write code → Understand problem → Algorithm
- D. Test → Compile → Understand problem → Design

27. What is the output of the following Python code?

```
a = 10
```

```
b = 5
```

```
print(a + b)
```

- A. 5
- B. 10
- C. 15
- D. 105

28. What will be the data type of x?

```
x = 25
```

- A. str
- B. float
- C. int
- D. bool

29. Which statement correctly takes input from the user in Python?

- A. scanf()
- B. input()
- C. read()
- D. get()

30. What will be the output?

a = 10

b = 3

print(a * b)

- A. 13
- B. 7
- C. 30
- D. 3.33

Answer any 4 | 2 marks each.

1. Define **Algorithm**. Write any **two characteristics** of an algorithm.
2. What is a **Flowchart**? Name any **four commonly used flowchart symbols** and state their purposes.
3. What is **Pseudocode**? How is it different from actual program code?
4. What is a **Compiler** and what is an **Interpreter**? Write one difference between them.
5. What is **Python**? Explain why Python is called a **high-level, general-purpose programming language**.

Part C — Programming Questions Any 3 | 4 marks each

Q1. Write a Python program that takes **two numbers from the user** and displays their:

- Sum
- Difference
- Product

Q2. Draw a flowchart to calculate the area of a rectangle.

The flowchart should:

- Take length and breadth as input.
- Calculate the area using:

Area = Length × Breadth

- Display the calculated area.
- End the flowchart.

Q3. Write a Python program to take **Principal, Rate and Time** from the user and calculate Simple Interest.

Formula:

$$SI = (P \times R \times T) / 100$$

```
print("Sum =", a + b)
```

This is the only dedicated typecasting question, as you requested.

Q4. Write a Python program that takes **three numbers from the user** and calculates their average.

Formula:

$$\text{Average} = (a + b + c) / 3$$